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SAP MM IMPLEMENTATION DOCUMENTATION

Procure-to-Pay (P2P) — Full Purchasing Cycle

NexGen Manufacturing Ltd.

SAP ECC 6.0 | Greenfield Implementation | SAP MM Module

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3. Business Requirements

This section documents the existing procurement landscape at NexGen Manufacturing Ltd. prior to SAP implementation (the AS-IS state), articulates the business rationale for adopting SAP MM, and defines the agreed scope of implementation.

3.1 Current Challenges (AS-IS Process)

Prior to SAP implementation, NexGen Manufacturing Ltd. operated its procurement processes through a combination of Microsoft Excel spreadsheets, email-based communication, and a legacy ERP system. This environment introduced significant operational inefficiencies and compliance risks. The key pain points identified during the AS-IS process assessment are summarized below:

Pain Point	Description
No Centralized Vendor Database	Each plant maintained its own vendor list independently, resulting in duplication and inconsistency across locations.
Manual Purchase Orders	Purchase Orders were created manually, making them prone to data entry errors and missing approval steps.
No 3-Way Matching	Vendor invoices were processed without verification against corresponding Purchase Orders and Goods Receipts.
Inventory Mismatches	Spreadsheet-based stock records frequently diverged from physical inventory, causing procurement over-ordering.
Delayed Approvals	PO approvals required physical signatures, causing delays of 5 to 7 business days per transaction.
No Spend Analysis	Management lacked real-time visibility into procurement expenditure by commodity, vendor, or plant.
Duplicate Payments	Absence of centralized tracking led to the same vendor invoice being processed multiple times.
Audit Failures	Insufficient documentation trail impaired audit readiness and regulatory compliance under internal SOX controls.

3.2 Why SAP Is Needed

SAP Materials Management (MM) directly resolves each of the above challenges through a structured, automated, and fully integrated procurement platform. The mapping between identified business problems and corresponding SAP MM solutions is detailed below:

Business Problem	SAP MM Solution
No centralized vendor database	Vendor Master in SAP — single consolidated record accessible across all plants (XK01).

Business Problem	SAP MM Solution
Manual Purchase Orders	SAP PO with configurable workflow-based approvals via Release Strategy (ME21N).
No 3-way matching	MIRO performs automatic PO–GR–Invoice matching before payment authorization.
Inventory mismatches	Real-time stock updates triggered automatically by MIGO goods receipt posting.
Delayed approvals	SAP Release Strategy with automated email notifications to designated approvers.
No spend analysis	SAP standard procurement reports: ME2M (by material), ME2N (by PO), MB52 (warehouse stock).
Duplicate payments	Built-in duplicate invoice check mechanism in MIRO prevents repeated payment.
Audit failures	Every SAP transaction carries a unique document number, creating a complete digital audit trail.

3.3 Scope of SAP Implementation

The boundaries of this SAP implementation engagement are formally defined as follows:

Scope Dimension	Detail
Module In Scope	SAP MM (Materials Management)
Integration In Scope	SAP FI — Accounts Payable (FI-AP)
Plant Locations Covered	Bhubaneswar (NX10), Pune (NX20), Chennai (NX30)
Process In Scope	Full Procure-to-Pay cycle — Purchase Requisition through Vendor Payment
Out of Scope (Phase 2)	SAP SD (Sales & Distribution), SAP HR, SAP PP (Production Planning)

4. SAP Module Overview — SAP MM

This section provides a formal overview of the SAP Materials Management (MM) module, its functional sub-components, and the end-to-end Procure-to-Pay cycle it supports within the NexGen Manufacturing environment.

4.1 What is SAP MM?

SAP MM (Materials Management) is a core functional module within the SAP ERP suite responsible for managing all procurement and inventory management processes across the enterprise. It forms an integral part of the SAP Logistics suite and operates in real-time integration with the following modules:

- SAP FI (Financial Accounting) — automatic posting of accounting entries upon goods receipt and invoice verification.
- SAP PP (Production Planning) — MRP-driven automatic Purchase Requisition generation based on BOM explosion.
- SAP SD (Sales & Distribution) — stock management linked to customer order fulfillment and delivery processing.

At NexGen Manufacturing Ltd., SAP MM is being implemented on SAP ECC 6.0 with a planned migration to SAP S/4HANA 2023. The implementation follows a Greenfield approach across all three plant locations under a centralized Purchasing Organization (PO01).

4.2 Key Components of SAP MM

SAP MM comprises several functionally distinct sub-modules, each addressing a specific domain within procurement and materials management:

Sub-Module	Description	Key T-Codes
Purchasing	Manages procurement activities: Purchase Requisitions, RFQs, Purchase Orders, Contracts, Scheduling Agreements.	`ME51N, ME41, ME21N`
Inventory Management	Handles goods receipts, goods issues, stock transfers, and physical inventory postings.	`MIGO, MB1A, MI01`
Invoice Verification (LIV)	Performs Logistics Invoice Verification via 3-way matching of PO, GR, and vendor invoice.	`MIRO, MIR4, MIR7`
Material Valuation	Manages material valuation using Standard Price or Moving Average Price methods.	`MR21, CKMVFM`
Vendor Evaluation	Scores and tracks vendor performance based on price, quality, on-time delivery, and service levels.	`ME61, ME6B`
Warehouse Management (WM)	Provides bin-level storage management integrated with Inventory Management for precise stock tracking.	`LT01, LP10`

4.3 P2P Cycle Overview in SAP MM

The Procure-to-Pay (P2P) cycle represents the complete lifecycle of a procurement transaction — from identification of a material requirement to final vendor payment. Within SAP MM, each stage produces a system-generated document with a unique reference number, creating a fully auditable and unbroken document chain. The seven-stage P2P cycle as configured for NexGen Manufacturing Ltd. is summarized below:

Step	Document / Activity	T-Code	Responsible Role
1	**Purchase Requisition (PR)**	ME51N	End User / Department
2	**Request for Quotation (RFQ)**	ME41	Purchasing Officer
3	**Quotation Received & Evaluated**	ME47 / ME49	Purchasing Officer
4	**Purchase Order (PO) Issued to Vendor**	ME21N	Purchasing Officer
5	**Goods Receipt (GR) Posted**	MIGO	Warehouse / Store Keeper
6	**Invoice Verification (3-Way Match)**	MIRO	Accounts Payable
7	**Vendor Payment Processed**	F110	Finance Team

Note: For service-based procurement (e.g., Annual Maintenance Contracts), Item Category D is used with Service Entry Sheets (T-Code: ML81N) in place of a standard Goods Receipt. All other steps remain identical to the standard P2P cycle described above.

5. Organizational Structure in SAP

The SAP organizational structure defines the legal, operational, and procurement hierarchy within the system. Correct configuration of organizational units is a prerequisite for all subsequent MM master data and transactional processing. This section describes the units defined for NexGen Manufacturing Ltd.

5.1 SAP Organizational Units Explained

Each SAP organizational unit corresponds to a specific level of the enterprise structure. The mapping between SAP organizational units and their NexGen-specific values is detailed in the reference table below:

SAP Org Unit	Definition	NexGen Value
Client	Highest level in SAP — represents the entire enterprise group.	100 (NexGen Group)
Company Code	Legal entity for financial accounting — generates P&L and Balance Sheet.	NX01 (NexGen Mfg. Ltd.)

SAP Org Unit	Definition	NexGen Value
Plant	Physical location where materials are stored, produced, or distributed.	`NX10, NX20, NX30`
Storage Location	Sub-division within a plant designated for specific categories of material storage.	`SL01, SL02, SL03`
Purchasing Organization	Organizational unit responsible for all procurement negotiations and PO issuance.	`PO01 (Central Purchasing)`
Purchasing Group	Buyer or group of buyers responsible for specific material categories.	`PG-RM, PG-SP, PG-SV`
Valuation Area	Level at which material stock value is determined (configured at Plant level at NexGen).	`Plant-level valuation`

5.2 Assignments in SAP

The following organizational assignments have been configured in the SAP Enterprise Structure (SPRO) for NexGen:

- Company Code NX01 assigned to Controlling Area CA01 for management accounting integration.
- All plants (NX10, NX20, NX30) assigned to Company Code NX01.
- Purchasing Organization PO01 assigned to Company Code NX01 (cross-company purchasing enabled).
- PO01 additionally assigned to each individual plant to enable plant-level PO creation.
- Purchasing Groups PG-RM, PG-SP, and PG-SV assigned to buyers within PO01.

5.3 Full Organizational Hierarchy

The complete enterprise hierarchy as configured in SAP for NexGen Manufacturing Ltd. is represented below:

```
CLIENT: 100 (NexGen Group)
  |
  +-- COMPANY CODE: NX01 (NexGen Manufacturing Ltd.)  |  Currency: INR  |
Country: IN
       |
       +-- PLANT: NX10 (Bhubaneswar)    SL01-Raw Materials | SL02-
Finished Goods | SL03-Spares
       +-- PLANT: NX20 (Pune)           SL01-Raw Materials | SL02-
Finished Goods
       +-- PLANT: NX30 (Chennai)        SL01-Raw Materials | SL02-
Finished Goods
       |
       +-- PURCHASING ORG: PO01 (Central — covers all plants)
              +-- PG-RM  (Raw Materials Buyers)
              +-- PG-SP  (Spare Parts Buyers)
              +-- PG-SV  (Services Buyers)
```

6. Master Data Configuration

Master data in SAP MM forms the foundation for all transactional processing. Incomplete or incorrect master data is one of the primary causes of implementation failures. This section covers the three critical master data objects configured for NexGen: Vendor Master, Material Master, and GL Accounts.

6.1 Vendor Master

The Vendor Master is the central repository of all supplier information in SAP. It is structured across three organizational levels, each maintained by a different team:

- General Data (Client Level) — Vendor name, address, communication details, bank account information.
- Company Code Data (NX01) — Reconciliation account, payment terms, withholding tax configuration.
- Purchasing Organization Data (PO01) — Incoterms, order currency, delivery and invoicing conditions.

Sample Vendor Master record configured for NexGen Manufacturing Ltd.:

Field	Value
Vendor Code	V-1001
Vendor Name	Tata Steel Supply Co. Pvt. Ltd.
City / Country	Jamshedpur, India
Reconciliation Account (FI)	160000 — Accounts Payable (Domestic)
Payment Terms	ZN30 — Net 30 days from invoice date
Order Currency	INR (Indian Rupee)
Incoterms	CIF — Cost, Insurance, and Freight
T-Code (Central)	XK01 (Create) | XK02 (Change) | XK03 (Display)

6.2 Material Master

The Material Master consolidates all technical and organizational data about a material. It is organized into views, with each department responsible for maintaining their relevant fields:

View	Maintained By	Key Fields
Basic Data 1 & 2	Master Data Team	Material description, base unit of measure, weight, volume, material group.
Purchasing View	Purchasing Team	Purchasing group, planned delivery time, order unit, over/under-delivery tolerance.
MRP 1 & 2	Production Planning	MRP type, lot size procedure, reorder point, safety stock, planned delivery time.
Accounting 1 & 2	Finance Team	Valuation class, price control (Standard/Moving Average), standard/moving avg price.
Storage 1 & 2	Warehouse Mgmt	Storage conditions, temperature requirements, shelf life, hazardous material indicator.
Quality Management	QM Team	Inspection types, quality certificate requirements, QM control key.

Sample Material Master record for NexGen's primary raw material:

Field	Value
Material Code	RM-5001
Description	Hot Rolled Steel Sheet 3mm
Material Type	ROH — Raw Material
Industry Sector	M — Mechanical Engineering
Base Unit of Measure	KG (Kilogram)
Valuation Class	3000 — Raw Materials
Price Control	V — Moving Average Price
MRP Type	PD — MRP Deterministic (Consumption-Based)
T-Code (Create)	MM01

6.3 GL Accounts — FI Integration

Every goods movement and invoice posting in SAP MM automatically triggers a financial accounting entry in FI. The GL account determination is configured via T-Code OBYC (Automatic Account Determination). The key GL accounts used across the P2P cycle at NexGen are:

GL Account	Account Description	Usage in P2P Cycle
300000	Raw Material Inventory	Debited when Goods Receipt (MIGO — Movement Type 101) is posted.

GL Account	Account Description	Usage in P2P Cycle
191000	GR/IR Clearing Account	Bridge account: credited at GR, debited at IR. Nets to zero after both postings.
160000	Accounts Payable — Domestic	Credited when vendor invoice is posted in MIRO; debited when payment is made via F110.
110000	Bank Account	Credited when outgoing payment to vendor is executed via F110.
476000	Purchase Price Variance	Records difference between PO price and actual invoice price during MIRO.

The GR/IR Clearing Account (GL 191000) acts as a temporary bridge between the procurement and finance functions. Upon goods receipt posting, GL 191000 is credited; upon invoice posting in MIRO, GL 191000 is debited. When both events are complete, this account nets to zero, signifying full and matched processing of the procurement cycle.